

## HURGHADA INTL, Egypt

WMO: 624630

Lat: 27.178N

Lon: 33.799E

Elev: 16

StdP: 101.14

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | 10.2       | 11.2 | -5.0                          | 2.5 | 17.7 | -3.1 | 2.9 | 17.9 | 12.8                  | 18.6 | 11.9 | 18.5 | 4.9                      | 290  | 0.596 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 9.0                    | 39.9            | 22.3 | 38.8 | 22.1 | 37.7 | 22.1 | 25.8                | 34.0 | 25.0 | 33.8 | 24.3 | 33.5 | 6.1                  | 0    |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 23.7                            | 18.5 | 30.4 | 22.2 | 16.9 | 30.4 | 21.2 | 15.9 | 30.5 | 79.9          | 33.7 | 76.3 | 33.7 | 73.5 | 33.7 | 30.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |     |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-----|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |     |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB  | WB  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |     |
| (4) 13.1          | 12.0 | 11.0 | 8.0                        | 42.2 | 1.1                | 1.9 | 7.2                                                | 43.6 | 6.5        | 44.7 | 5.9        | 45.7 | 5.1        | 47.1 |     |     |
| (5)               |      |      | 3.3                        | 27.6 | 0.8                | 1.5 | 2.7                                                | 28.7 | 2.2        | 29.5 | 1.8        | 30.3 | 1.2        | 31.4 | (4) | (5) |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 25.5   | 17.0  | 18.2  | 20.9  | 24.4  | 28.5  | 31.3  | 32.8  | 32.8  | 30.5  | 27.1  | 22.7  | 18.7 |
|                                                                               | DBStd       | 6.01   | 1.94  | 2.30  | 2.67  | 2.74  | 2.52  | 2.09  | 1.61  | 1.53  | 1.80  | 2.13  | 2.23  | 1.94 |
|                                                                               | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | HDD18.3     | 102    | 51    | 27    | 6     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 18   |
|                                                                               | CDD10.0     | 5641   | 217   | 231   | 338   | 431   | 575   | 638   | 706   | 708   | 616   | 530   | 382   | 270  |
|                                                                               | CDD18.3     | 2701   | 9     | 24    | 86    | 181   | 317   | 388   | 447   | 449   | 366   | 272   | 133   | 29   |
|                                                                               | CDH23.3     | 36076  | 65    | 179   | 617   | 1688  | 4099  | 5898  | 7102  | 7181  | 5283  | 2931  | 882   | 153  |
|                                                                               | CDH26.7     | 20302  | 4     | 36    | 163   | 621   | 2094  | 3573  | 4628  | 4705  | 2989  | 1246  | 221   | 22   |
| Wind                                                                          | WSAvg       | 6.0    | 5.3   | 5.5   | 5.8   | 5.8   | 6.1   | 7.0   | 6.3   | 6.8   | 6.8   | 5.8   | 5.3   | 5.3  |
| Precipitation                                                                 | PrecAvg     | 8      | 1     | 0     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 2     | 1    |
|                                                                               | PrecMax     | 61     | 4     | 2     | 5     | 0     | 8     | 0     | 0     | 0     | 0     | 49    | 60    | 15   |
|                                                                               | PrecMin     | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | PrecStd     | 15     | 1     | 0     | 1     | 0     | 1     | 0     | 0     | 0     | 0     | 9     | 11    | 4    |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 27.1  | 30.1  | 33.1  | 36.2  | 40.2  | 41.8  | 41.2  | 41.0  | 39.3  | 36.2  | 32.2  | 29.1 |
|                                                                               |             | MCWB   | 17.8  | 17.1  | 18.4  | 18.6  | 20.6  | 21.4  | 22.8  | 22.9  | 22.8  | 22.5  | 20.2  | 19.2 |
|                                                                               | 2%          | DB     | 25.1  | 27.2  | 30.3  | 33.9  | 37.1  | 39.0  | 39.4  | 39.2  | 37.6  | 34.5  | 30.4  | 26.4 |
|                                                                               |             | MCWB   | 17.0  | 17.0  | 17.7  | 18.7  | 20.0  | 21.5  | 22.4  | 22.8  | 22.5  | 22.0  | 20.5  | 17.7 |
|                                                                               | 5%          | DB     | 23.4  | 25.2  | 28.1  | 31.9  | 35.2  | 37.4  | 38.1  | 38.1  | 36.2  | 33.1  | 29.1  | 24.9 |
|                                                                               |             | MCWB   | 15.8  | 16.3  | 17.5  | 18.0  | 19.9  | 21.2  | 22.3  | 22.8  | 22.4  | 21.8  | 20.2  | 16.8 |
|                                                                               | 10%         | DB     | 22.0  | 23.8  | 26.4  | 30.0  | 33.9  | 36.1  | 37.1  | 37.1  | 35.0  | 31.9  | 27.8  | 23.7 |
|                                                                               |             | MCWB   | 14.4  | 15.6  | 16.8  | 17.8  | 19.6  | 21.1  | 22.2  | 22.7  | 22.3  | 21.7  | 19.2  | 15.8 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 21.7  | 22.6  | 23.7  | 24.2  | 24.6  | 24.7  | 26.4  | 27.3  | 26.6  | 26.1  | 24.7  | 22.0 |
|                                                                               |             | MCDB   | 24.9  | 24.9  | 26.6  | 27.6  | 30.7  | 34.6  | 34.2  | 35.0  | 34.2  | 31.6  | 28.4  | 26.0 |
|                                                                               | 2%          | WB     | 18.6  | 20.0  | 21.5  | 22.3  | 23.0  | 23.6  | 25.2  | 25.8  | 25.1  | 24.7  | 22.8  | 19.4 |
|                                                                               |             | MCDB   | 23.1  | 24.5  | 25.6  | 27.2  | 30.7  | 34.5  | 34.6  | 35.1  | 33.6  | 31.1  | 27.8  | 24.6 |
|                                                                               | 5%          | WB     | 16.5  | 17.8  | 19.6  | 20.9  | 21.8  | 22.7  | 24.4  | 24.9  | 24.2  | 23.6  | 21.3  | 17.5 |
|                                                                               |             | MCDB   | 22.5  | 24.1  | 25.8  | 27.8  | 31.5  | 34.5  | 34.6  | 35.1  | 33.2  | 30.6  | 27.4  | 23.8 |
|                                                                               | 10%         | WB     | 15.0  | 15.9  | 17.9  | 19.3  | 20.8  | 22.1  | 23.7  | 24.2  | 23.5  | 22.6  | 20.0  | 16.2 |
|                                                                               |             | MCDB   | 21.3  | 23.3  | 25.3  | 28.3  | 31.8  | 34.1  | 34.5  | 34.6  | 33.0  | 30.2  | 26.8  | 22.7 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 10.2  | 10.5  | 10.7  | 10.6  | 10.2  | 9.5   | 9.1   | 9.0   | 9.2   | 9.3   | 9.8   | 10.0 |
|                                                                               |             | MCDBR  | 12.3  | 13.1  | 12.8  | 13.1  | 12.0  | 11.1  | 10.2  | 10.0  | 10.4  | 10.4  | 11.0  | 11.8 |
|                                                                               |             | MCWBR  | 8.5   | 9.0   | 8.8   | 8.2   | 7.0   | 6.4   | 6.3   | 6.3   | 6.3   | 6.9   | 7.5   | 7.8  |
|                                                                               | 5% WB       | MCDBR  | 11.8  | 12.4  | 11.8  | 11.4  | 10.6  | 10.0  | 9.3   | 9.1   | 9.3   | 9.5   | 10.4  | 11.3 |
|                                                                               |             | MCWBR  | 9.2   | 9.9   | 9.8   | 8.9   | 7.7   | 6.5   | 6.9   | 6.8   | 6.6   | 7.0   | 7.9   | 8.3  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.371  | 0.406 | 0.454 | 0.523 | 0.532 | 0.452 | 0.464 | 0.469 | 0.458 | 0.462 | 0.393 | 0.369 |      |
|                                                                               | taud        | 2.300  | 2.168 | 2.022 | 1.860 | 1.861 | 2.100 | 2.100 | 2.091 | 2.103 | 2.075 | 2.257 | 2.316 |      |
|                                                                               | Ebn at Noon | 866    | 861   | 839   | 791   | 782   | 842   | 830   | 826   | 824   | 793   | 833   | 849   |      |
|                                                                               | Edh at Noon | 113    | 140   | 171   | 207   | 207   | 162   | 162   | 162   | 156   | 151   | 117   | 107   |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 4.19   | 5.11  | 6.35  | 7.11  | 7.61  | 8.04  | 7.78  | 7.33  | 6.58  | 5.45  | 4.47  | 3.93  |      |
|                                                                               | RadStd      | 0.18   | 0.21  | 0.18  | 0.17  | 0.24  | 0.09  | 0.12  | 0.13  | 0.16  | 0.15  | 0.12  | 0.08  |      |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           |  | +0.57 | N/A     | N/A    | +0.94   | +0.68 | N/A   | N/A         | N/A     | +235    | +242    |
| (47) Regional (5 neighbors) |  | +0.57 | N/A     | N/A    | +0.57   | +0.53 | +0.54 | N/A         | -38     | +248    | +223    |

Nomenclature: See separate page